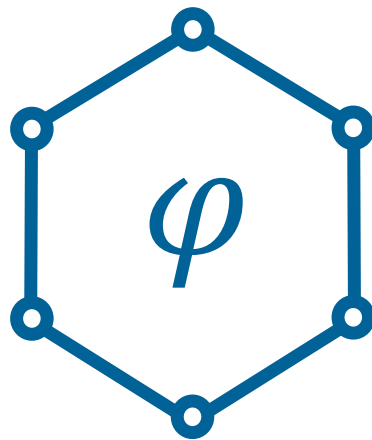
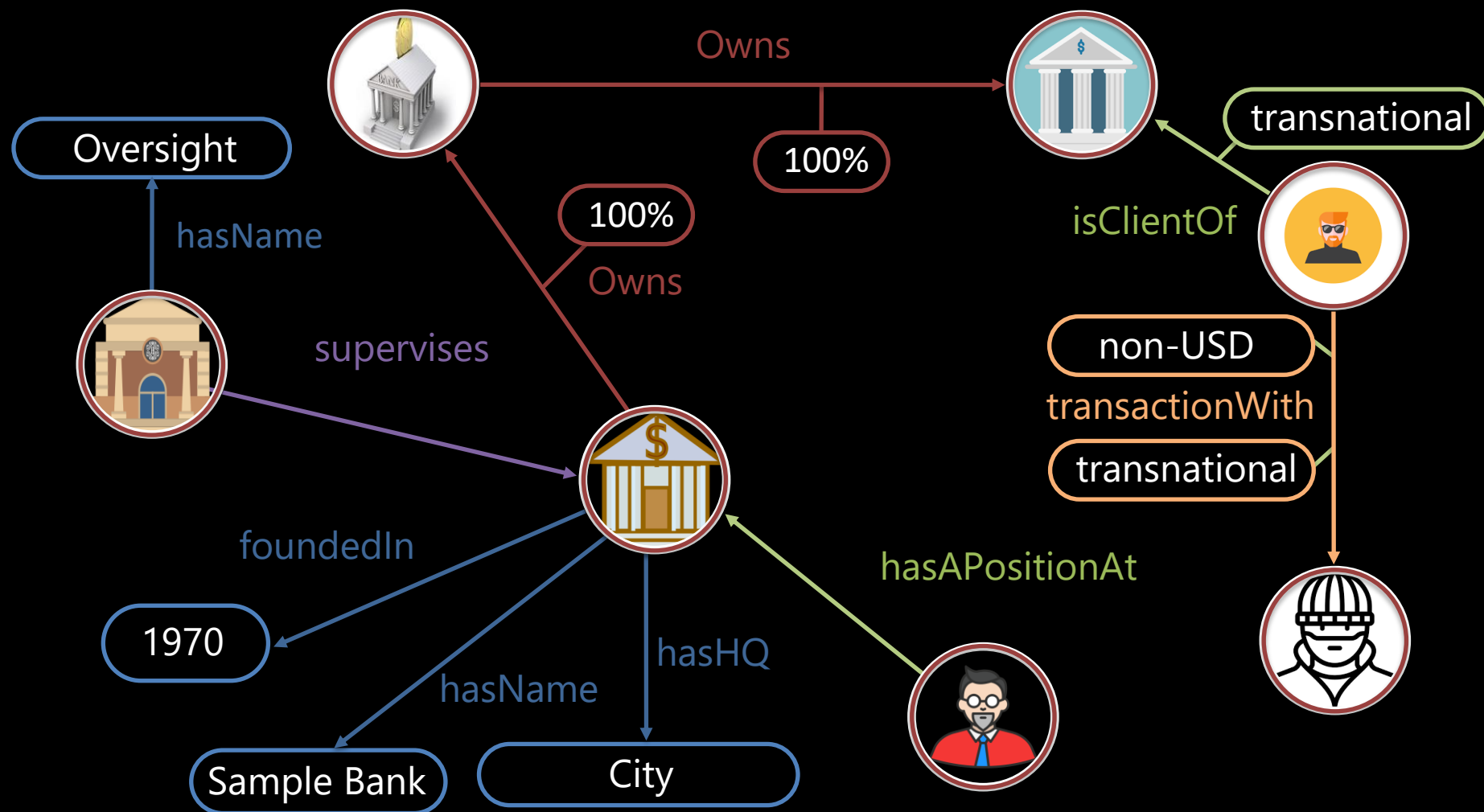
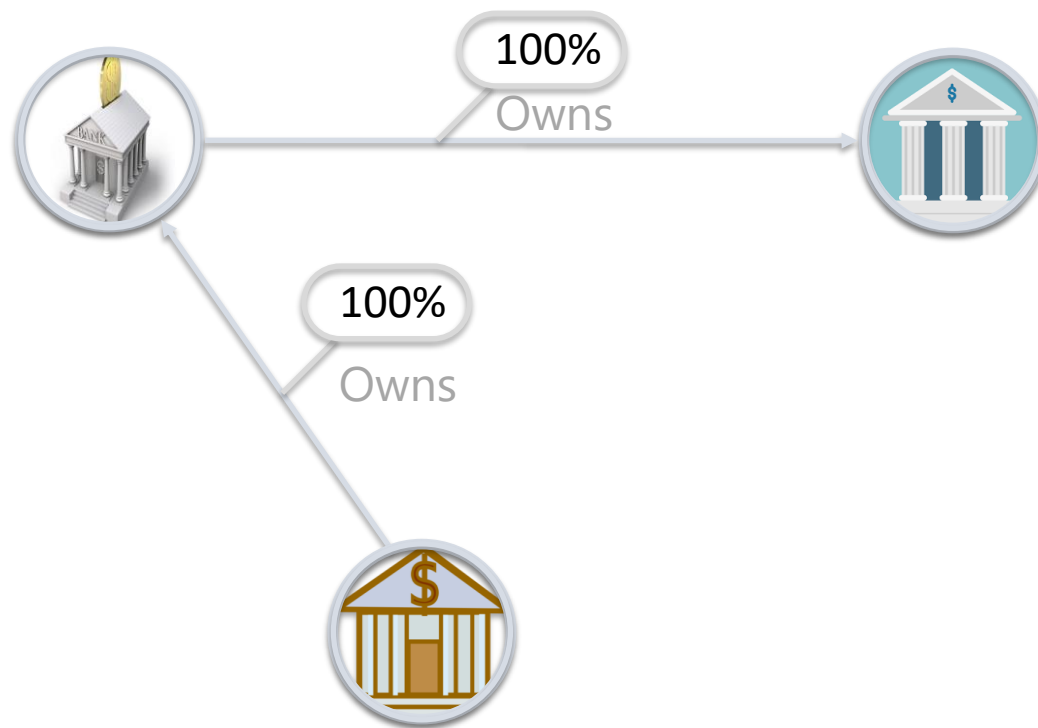


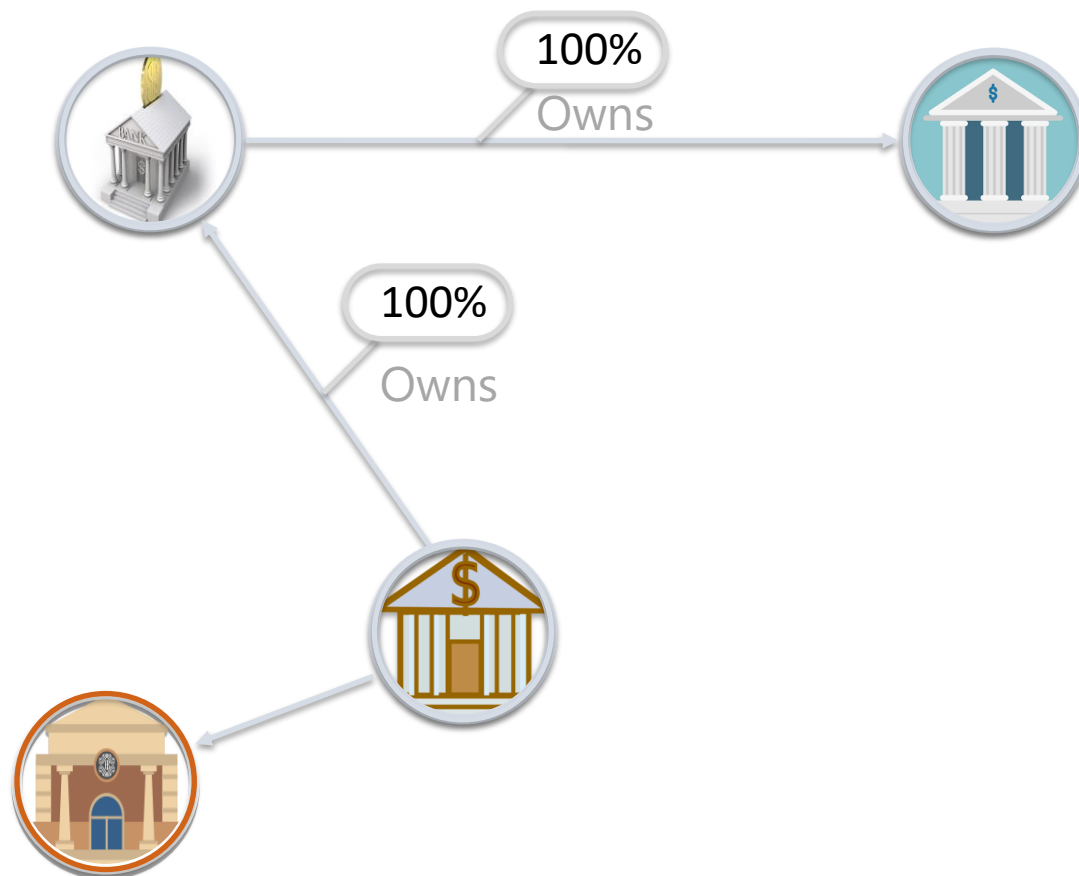


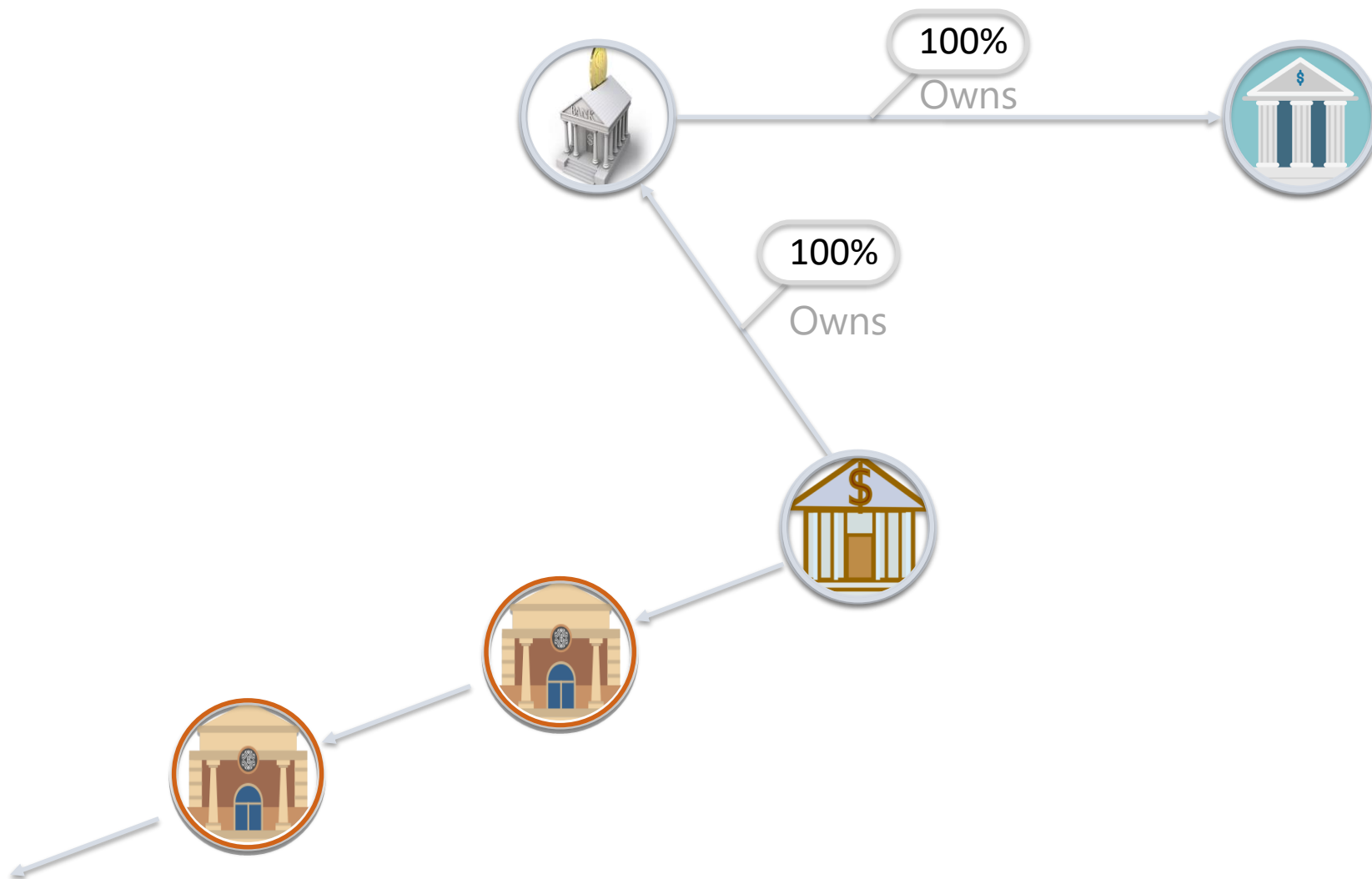
Logical Knowledge in KGs Creation

Emanuel Sallinger











PLC(HSBC, UK, London) → Company(HSBC)



$$\forall x, z \text{ PLC}(x, \text{UK}, z) \rightarrow \text{Company}(x)$$



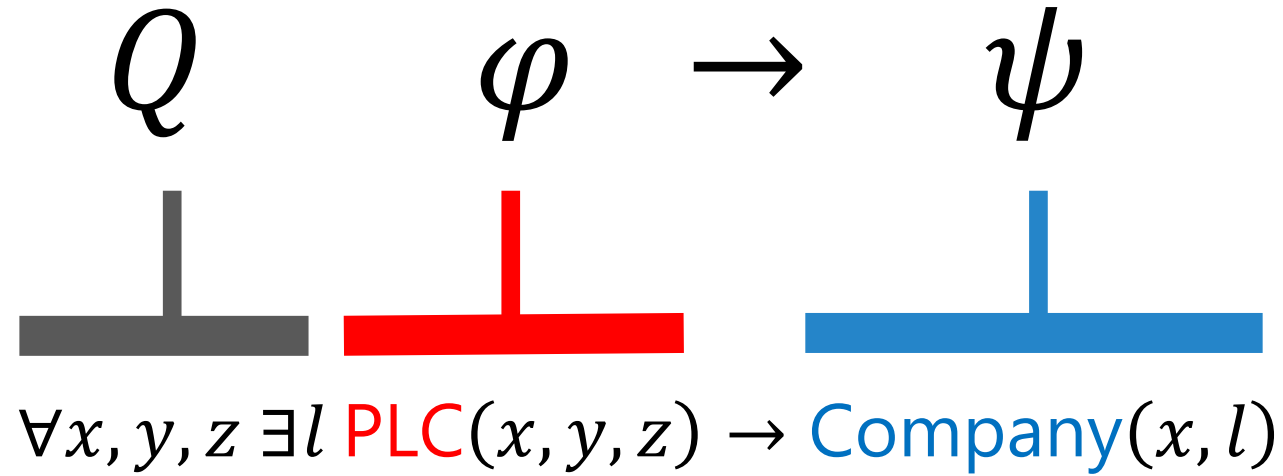
$$\forall x, y, z \text{ PLC}(x, y, z) \rightarrow \text{Company}(x)$$



$$\forall x, y, z \text{ PLC}(x, y, z) \rightarrow \exists l \text{ Company}(x, l)$$



$$\begin{array}{ccc} Q & \varphi & \rightarrow \psi \\ \text{---} & \text{---} & \text{---} \\ \forall x, y, z \exists l \text{ PLC}(x, y, z) & \rightarrow & \text{Company}(x, l) \end{array}$$



Knowledge

"Dependency"

"Rule"

"Logical Formula"

"Ontology **Axiom**"

"Constraint"

"Named **Query**"



SQL (PostgreSQL)

```
CREATE SEQUENCE locationSequence;  
  
SELECT id, nextval('locationSequence')  
INTO Company FROM PLC
```

```
Company(C,L) :- PLC(X,Y,C).
```



SQL: **NULL**

v IS [NOT] NULL

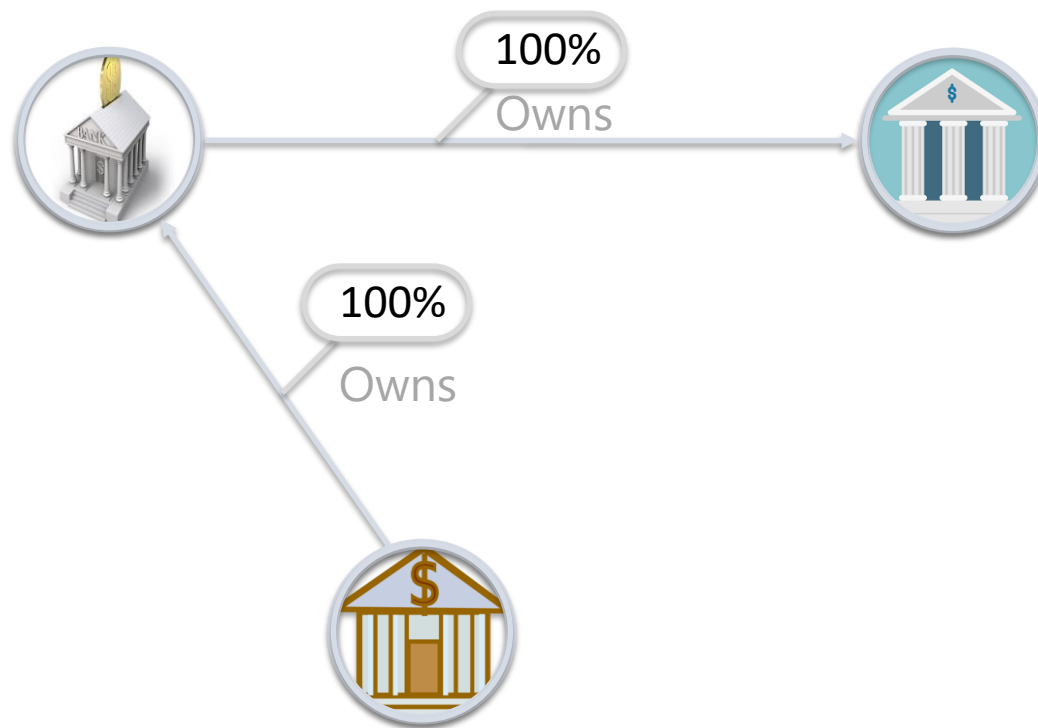
- **NULL** indicates an “unknown” value
 - many applications (not applicable, not existent, ...)
 - often misused
- **NULL** is different from any other value, including itself
 - **NULL = NULL** is not true!
 - to test for null, use **IS [NOT] NULL**

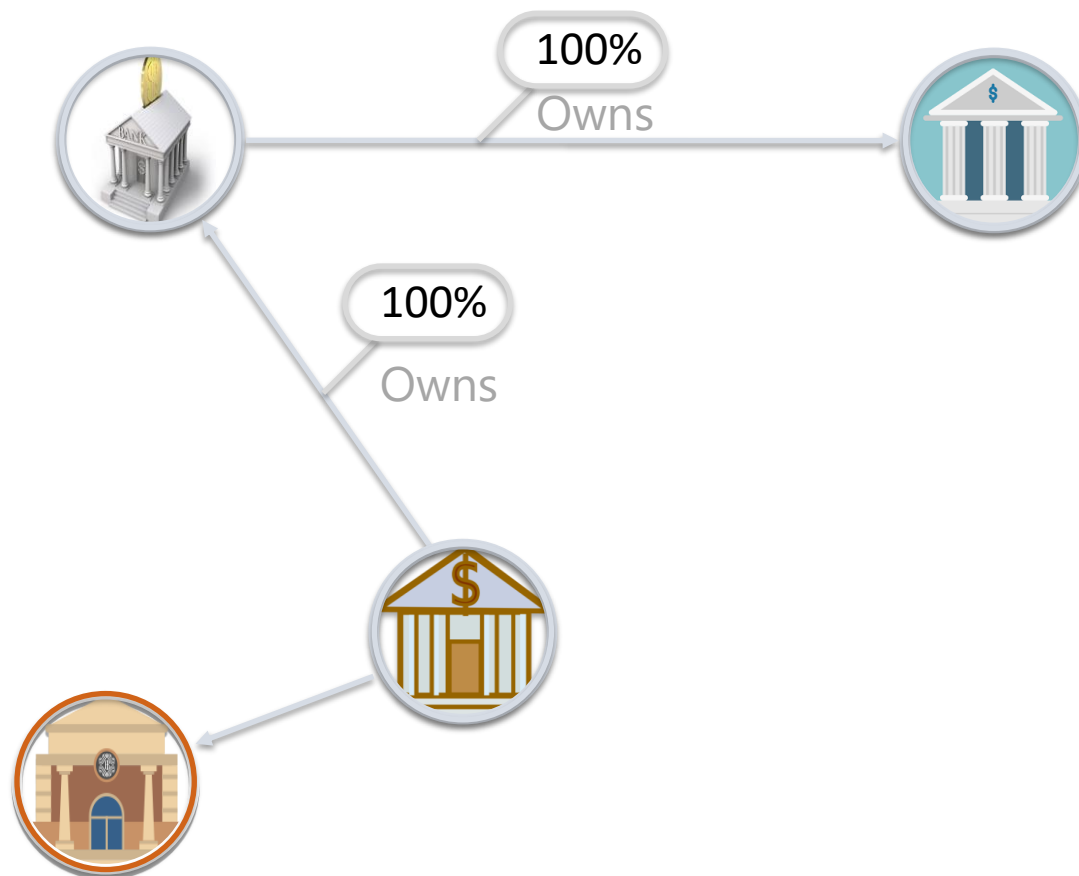


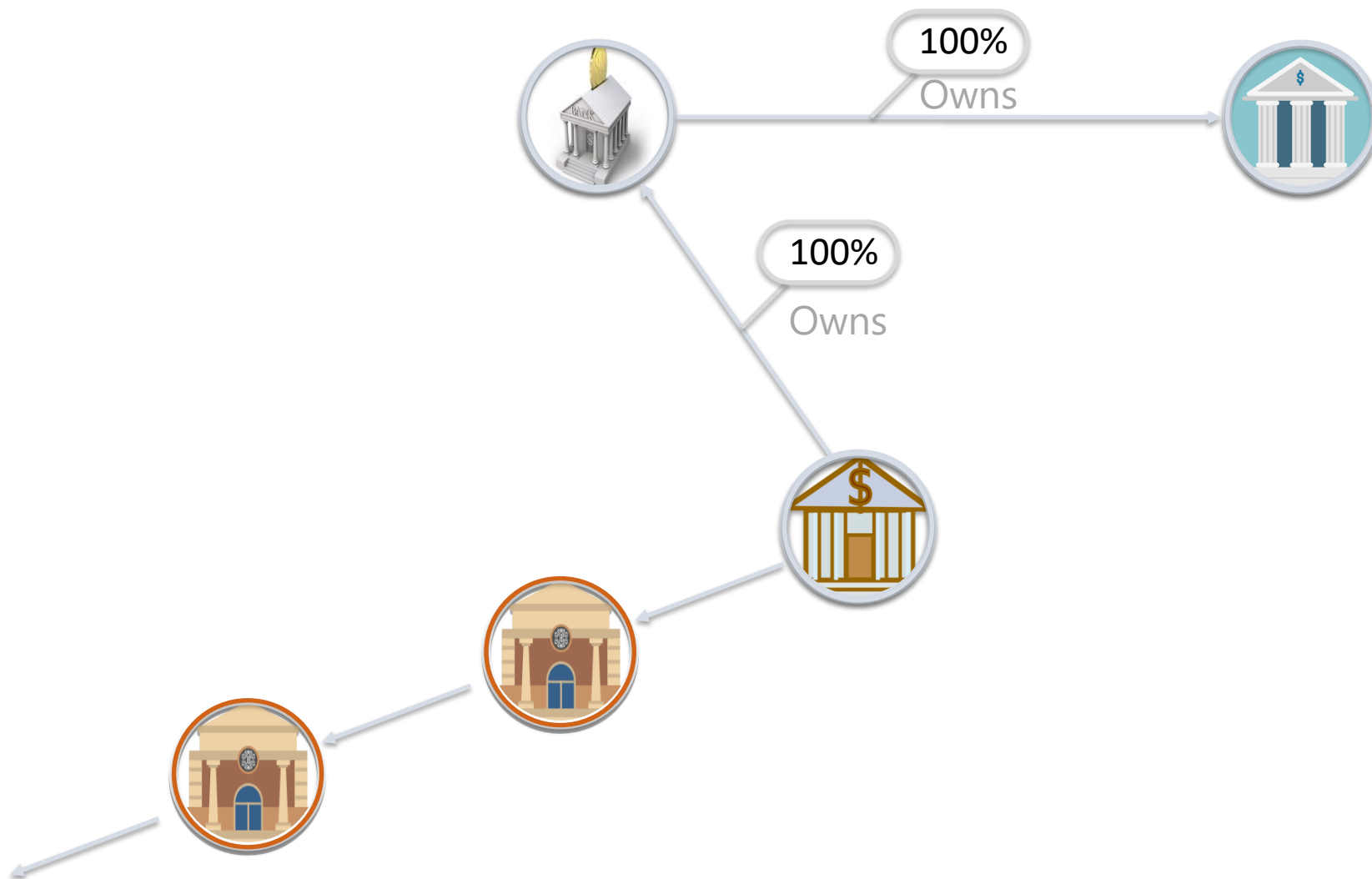
SQL: NULL

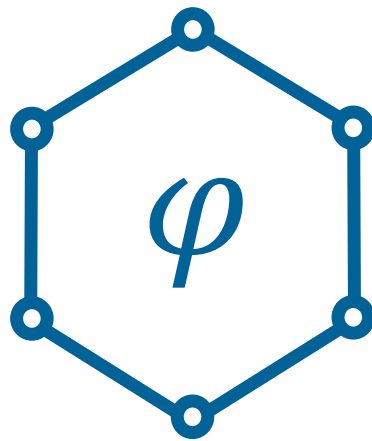
Recall

φ	ψ	φ AND ψ	φ OR ψ	NOT φ
true	true	true	true	false
true	false	false	true	false
true	null	null	true	false
false	true	false	true	true
false	false	false	false	true
false	null	false	null	true
null	true	null	true	null
null	false	false	null	null
null	null	null	null	null









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